

NAMRATA VISHWAKARMA

Bioengineering | Molecular Biology | Bioinformatics | Biomedical Research

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SUMMARY

Results-driven third-year B.Tech Bioengineering student at VIT Bhopal with hands-on experience in molecular biology, bioinformatics, and advanced microscopy. Passionate about bridging wet-lab research with computational biology to develop scalable, cost-effective biotechnological solutions. Seeking internship or research opportunities in biomedical R&D, biologics, or clinical diagnostics.

CORE COMPETENCIES

Molecular Biology · Microbial Culture · Gel Electrophoresis · PCR / qPCR · SEM & AFM Operation · Nanoparticle Characterization

Bioinformatics (Sequence Alignment, Comparative Genomics) · Python Scripting · GMP/GLP Compliance · Protein Expression · Scientific Writing

WORK EXPERIENCE & INTERSHIPS

Electron & Probe Microscopy Workshop — *Parul University, Gujarat*

Feb 25–27, 2026

- ▶ Operated Scanning Electron Microscope (SEM) and Atomic Force Microscope (AFM) for surface characterisation, supporting biosimilar and protein purification QC workflows.
- ▶ Performed nanoscale structural analysis and AFM image interpretation to assess nanoparticle morphology for downstream biopharmaceutical processing.
- ▶ Obtained formal certification in advanced microscopy techniques, strengthening analytical credibility for biomedical manufacturing roles.

Bioinformatics Internship — *BioTecNika (Remote)*

Jun – Jul 2025

- ▶ Identified 3 novel candidate enzyme sequences for protein expression through computational screening and comparative genomics analysis.
- ▶ Integrated cell culture datasets with sequence analysis pipelines to support biologics R&D and upstream bioprocessing workflows.

Bioinformatics Skills Workshop — *DECode Life (Remote)*

May – Jun 2025

- ▶ Mastered end-to-end genomic data pipelines including sequence alignment, variant calling, and PCR/qPCR data interpretation for independent dataset analysis.
- ▶ Produced 4+ case studies using GLP-aligned data protocols, mirroring industry-standard biopharmaceutical quality-assurance workflows.
- ▶ Automated sequence processing tasks using Python scripting, achieving a **40% reduction** in manual annotation time and improving reproducibility.

RESEARCH PROJECTS

Nanoparticles Promoting Plant-Microbe Symbiosis — *VIT Bhopal*

2024

- ▶ Investigated the mechanistic influence of nanoparticles on microbial rhizosphere interactions to enhance crop yield and plant growth through targeted experimental design.
- ▶ Applied principles of nanobiotechnology and microbiology, generating data relevant to sustainable agricultural biotech applications.

Plant X: Biointelligent Plastic Degradation — *VIT Bhopal*

2024

- ▶ Conceptualised and designed a synthetic biology solution utilising engineered microorganisms for household plastic (PET/HDPE) biodegradation.
- ▶ Developed a scalable microbial chassis model integrating gene circuit design for real-world environmental application, presented at institutional level.

TECHNICAL SKILLS

Laboratory Techniques: Microbial culture & maintenance, Gel electrophoresis, PCR/qPCR, SEM operation, AFM image acquisition & analysis, Nanoparticle characterisation, Cell culture (basic)

Bioinformatics & Computing: Sequence alignment (BLAST, ClustalW), Comparative genomics, Python (scripting, automation), R (data analysis), NCBI/UniProt databases, Linux/command-line basics

Regulatory & Quality: audit-ready record-keeping, data traceability

Soft Skills: Scientific writing, Research communication, Problem-solving, Team collaboration, Time management

EDUCATION

B.Tech in Bioengineering — *Vellore Institute of Technology (VIT), Bhopal*

2023 – 2027

- ▶ **Relevant Coursework:** Molecular Biology, Biochemistry, Bioprocess Engineering, Genomics, Bioinformatics, Microbiology, Biomaterials

CERTIFICATIONS & TRAINING

- ▶ Advanced Microscopy Certification (SEM & AFM) — Parul University, Gujarat *Feb 2026*
- ▶ Bioinformatics Skills Workshop Certificate — DECode Life *Jun 2025*
- ▶ Bioinformatics Internship Certificate — BioTecNika *Jul 2025*

EXTRACURRICULAR & ACADEMIC ACTIVITIES

- ▶ **Poster Presentation** — International Conference on Sustainable Aerospace Technology (interdisciplinary science communication).
- ▶ **Hackathon Participant** — VIT Solvit (Problem-Solving Track) and BHARAT GEN AI (AI-Based Biotech Solutions).
- ▶ Active member of science outreach, genomics research, and sustainability workshops at VIT Bhopal.